Lunar Module

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Developed to implement the 1962 lunar orbit rendezvous decision, the lunar module was a two stage, self-sufficient spacecraft built by Grumman. It was carried to the Moon attached to the command module docking port, with its powerful descent engine pointing forward. In this configuration, the LM descent engine could be fired as a back-up propulsion system to boost the Apollo-LM combination out of lunar orbit in the event the Apollo engine failed before the landing sequence.

The LM stood on four cantilevered landing legs which looked insect-like while retracted during flight. Each of the four legs ended in a dish-shaped pod which Grumman had designed to support the vehicle on a variety of surfaces. One leg was equipped with a ladder. The landing legs were attached to the descent stage which carried a throttleable rocket engine. On the Moon, the descent stage served as a launching platform for the ascent stage, which consisted of a pressurized cabin, guidance and navigation systems, including radar altimeter and computer, and a throttleable rocket engine. While it was docked with Apollo, the crew entered or left it through a tunnel. On the Moon, crewmen depressurized the cabin, opened the hatch and climbed out on to a small "porch" and then down the descent stage ladder to the lunar surface.

When the crew completed outside surface activites, they climbed back into the ascent stage, hoisted samples aboard with line and pulley, closed the hatch, re-pressurized the cabin and lifted off the descent stage by firing the ascent engine. The ascent stage would rendezvous and dock with the Apollo in lunar orbit. The crewmen and their samples would then transfer to the command module. After safely returning to Apollo, the ascent stage was jettisoned.

The odd shape of this spacecraft was dictated by the fact that it operated only in airless space; it did not have to be streamlined in any way. Two flight stations, at which the astronauts stood with harness restraints, had display panels, armrests, controls and landing aids, two front windows, and an overhead docking window. The landing leg struts, released explosively, were extended by springs. The main struts were filled with crushable aluminum honeycomb for absorbing compressive loads. All except the forward one were fitted with 1.7 m long sensing probes which upon contact with the lunar surface signalled the crew to shut down the descent engine.

For more information about the Apollo Moon program, see Apollo.

LUNAR MODULE SPECIFICATIONS

First flight: 22-Jan-1968; first manned flight 3-Mar-1969 (Apollo 9)
Last flight: 7-Dec-1972 (Apollo 17)
Number of manned flights: 9
Principal uses: manned lunar landing
Unit cost: $50.00 million
Crew size: 2
Overall height: 7.0 m
Maximum diameter: 4.3 m / 9.4 m (diagonally across landing legs)
Habitable volume: 4.5 m^3
Total mass: 15,200 kg for H-series (16,440 kg for J-series)
Propellant mass: 10,730 kg total (11,250 kg)
Primary engine thrust: 43.90 kN
Main engine propellant: NTO/Aerozine-50
Total spacecraft delta v: 4,690 m/s
Power: batteries; 65 kWh total (77 kWh)
Contractor: Grumman Aerospace Corp.

(Masses are not typical for every mission, figures given are median.)

ASCENT STAGE
The Lunar Module operated as a unit until the time of leaving the Moon when the ascent stage (which contained the crew) functioned as a separate spacecraft for rendezvous and docking with the Command Module. This section had a docking port for linking up with the command ship and crew transfer.
Crew size: 2
Height: 3.2 m
Maximum diameter: 4.3 m
Crew compartment: 2.35 m diameter x 1.07 m long
Habitable volume: 4.5 m^3
Total mass: 4,780 kg
Crew mass: 144 kg
Propellant mass: 2,375 kg
Reaction control system
  thrusters: 16 x 445 N
  propellant: NTO/Aerozine-50
  specific impulse: 290 s
Ascent engine
  thrust: 15.57 kN
  propellant: NTO/Aerozine-50
  specific impulse: 311 s
  delta v: 2,220 m/s
Power: Ag-Zn batteries; 2 × 296 Ah each, 28 V DC; 17 kWh; inverters produced 115 V AC

DESCENT STAGE
Height: 3.8 m
Maximum diameter: 4.3 m / 9.4 m (diagonally across landing legs)
Total mass: 10,465 kg for H-series (J-series: 11,665 kg)
Propellant mass: 8,355 kg (8,873 kg)
Descent engine
  thrust: throttleable 4.67-43.90 kN
  propellant: NTO/Aerozine-50
  specific impulse: 311 s
  delta v: 2,470 m/s
Power: Ag-Zn batteries; 4 × 415 Ah each, 28 V DC; 48 kWh for H-series (5 × 415 Ah, 60 kWh J-series)
Descent trajectory (min:s)
  00:00 - start descent; altitude 15.2 km, velocity 1,700 m/s, thrust 100%
  06:00 - throttle down; thrust 55%
  08:30 - pitchover; altitude 2.3 km
  12:30 - landing; velocity at touchdown 1 m/s

LUNAR ROVING VEHICLE
The LRV greatly extended the area of the lunar surface which could be explored by astronauts of the Apollo 15, 16 and 17 missions. Deceptive in appearance it looked like a simple "dune buggy" but in fact was a specialized space vehicle designed to operate in conditions of vacuum, wide extremes of temperature and over difficult terrain. To keep weight to a minimum the electrically powered vehicle was built largely of aluminum. Its specially woven wire wheels were made of coated piano wire. The LRV folded for stowage in the descent stage in the LM in a quadrant to the right of the ladder. The chassis was hinged in three places and the four wheels were pivoted nearly flat against the folded chassis occupying only 0.85 m^3.
Length: 3.1 m
Width: 1.8 m (to center of wheels)
Wheel base: 2.3 m
Wheel diameter: 81.3 cm
Ground clearance: 35.5 cm
Turning radius: 3.05 m
Maximum speed: 14 km/hr
Power supply: two 36-volt primary silver-zinc batteries
Traction drive: four 1/4 hp DC series wound motors (one each wheel)
Mass: 210 kg
Total loaded mass: 726 kg
Payload capacity: 490 kg

J-SERIES MODIFICATIONS
Descent Stage: Increased supplies of oxygen and water; extended electrical life. Quad 1 re-arranged to permit stowage of folded Lunar Roving Vehicle (LRV). Quad 4 - new 50 kg water tank, waste container and additional oxygen pressure tank and gaseous oxygen module replaces Modularised Equipment Stowage Assembly (MESA). A re-designed MESA is fitted outside Quad 4, includes tool pallet, sample containers, batteries for personnel life support systems, and cosmic ray detector. Four descent engine propellant tanks lengthened by 8.6 cm providing 521.5 kg extra fuel and oxidizer. Descent engine burned for longer period; combustion chamber modified to reduce erosion; expansion skirt modified.
Ascent Stage: New stowage for re-designed pressure suits of greater flexibility which had increased supplies of oxygen, water and electricity for extended EVAs. Plumbing added to waste container in descent stage.

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